# MUGDHA KHEDKAR

Secure Software Engineering, Heinz Nixdorf Institut, Universität Paderborn, Germany

https://mugdhakhedkar.com mugdha.khedkar@upb.de Google Scholar LinkedIn Orcid

## EDUCATION

**Doctorate in Computer Science (Dr. rer. nat.)** *September 2026*
Universität Paderborn, Germany
Advisor: Prof. Dr. Eric Bodden

**M.Sc. in Computer Science** *August 2020*
Chennai Mathematical Institute, India, CGPA: 8.25/10

**B.E. in Computer Engineering** *May 2017*
Cummins College of Engineering, Pune (India), 73%

## RESEARCH INTERESTS

Data Privacy and Protection, Software Security, Static Program Analysis, Empirical Software Engineering

## PUBLICATIONS

- **Mugdha Khedkar**. 2026. *Assisting GDPR Compliance through Static Analysis of Android Apps.* PhD Thesis, Paderborn University.
- **Mugdha Khedkar**, Michael Schlichtig, Mohamed Soliman & Eric Bodden. *Challenges in Android Data Disclosure: An Empirical Study*. In Proceedings of the IEEE/ACM 13th International Conference on Mobile Software Engineering and Systems (MOBILESoft'26). *(To appear)*
- **Mugdha Khedkar**, Michael Schlichtig, Nihad Atakishiyev & Eric Bodden. *Between Law and Code: Challenges and Opportunities for Automating Privacy Assessments.* Automated Software Engineering 33, 56 (2026).
- **Mugdha Khedkar**, Ambuj Kumar Mondal & Eric Bodden. *A study of privacy-related data collected by Android apps.* Automated Software Engineering 33, 45 (2026).
- **Mugdha Khedkar**, Michael Schlichtig & Eric Bodden. *Source Code-Driven GDPR Documentation: Supporting RoPA with Assessor View.* 2026 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER).
- **Mugdha Khedkar**, Ambuj Kumar Mondal & Eric Bodden. *Do Android App Developers Accurately Report Collection of Privacy-Related Data?* In Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering Workshops (ASEW'24).
- **Mugdha Khedkar**, Michael Schlichtig & Eric Bodden. *Advancing Android Privacy Assessments with Automation.* In Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering Workshops (ASEW'24).
- **Mugdha Khedkar** & Eric Bodden. *Toward an Android Static Analysis Approach for Data Protection.* In Proceedings of the IEEE/ACM 11th International Conference on Mobile Software Engineering and Systems (MOBILESoft'24).
- Mugdha Khedkar. *Static Analysis for Android GDPR Compliance Assurance.* 2023 IEEE/ACM 45th International Conference on Software Engineering Companion Proceedings (ICSE-Companion).

## INVITED TALKS AND WORKSHOPS (SELECTED)

- *Static Analysis for Privacy-Aware Reporting*, invited talk hosted by Prof. Tapti Palit, Davis Security Lab, University of California Davis. *Dec 2025*
- *From Pain Points to Automation: Support for Privacy Assessments*, invited workshop European Privacy KnowledgeNet (IAPP), Munich. *Oct 2025*
- *Static Analysis for Android GDPR Compliance*, invited talk hosted by Prof. Martina Lindorfer, Cybersecurity Center, TU Wien. *Oct 2025*

## SERVICES

- Program Committee Member: ICSE 2027 Student Research Competition, QUATIC 2026 (Thematic Track Security and Privacy), STATIC 2026 (co-located with ICSE), ASE 2026 Tools and Datasets, ASE 2025 and ASE 2024 Tool Demo

- Artifact Evaluation Committee Member: ASE 2026, USENIX 2026, PoPETS 2026
- Journal Reviewer: IEEE Transactions on Software Engineering, International Journal of Human-Computer Interaction, Information and Software Technology, Journal of Software: Evolution and Process, IEEE Transactions on Privacy, IEEE Software, Transportation Research Interdisciplinary Perspectives
- Junior Program Committee Member: MSR 2023 Technical Track
- Mentor: Universität Paderborn perspeEktIve M program *(Nov 2021 - Apr 2022)*
- Student Volunteer: ASE 2024, ASE 2021
- Participant: 4th Summer School on Security Testing & Verification 2025, Dagstuhl Research Methods Seminar 2023, ICSE 2022 Student Mentoring Workshop, SIGPLAN-M *(2021 – 2022)*, PLDI 2021 Mentoring Workshop

## MASTER THESIS SUPERVISION – Universität Paderborn

- Exploring Hybrid Analysis for Android Privacy *Ongoing*
- Automated Privacy Assessment for Android Applications: A DPV Visualization Approach *March 2025*
- Explaining Privacy-Relevant Program Slices to App Developers *Jan 2025*
- Automatically Generating Data Safety Section Labels for Android Apps *Jan 2025*
- Improving User Experience of Program Slicing wrt Data Privacy *Dec 2023*
- Labeling User-Input Privacy Data in Android Applications *July 2023*
- Empirical Evaluation of Forward and Backward Taint Analysis *June 2023*
- Qualitative Study of Data Flow Analysis Frameworks: IFDS, IDE and VASCO *Jan 2023*

## TEACHING EXPERIENCE – Universität Paderborn

- Secure Software Engineering (Bachelor course) *Summer 2025, Summer 2024*
- Seminar Secure Systems Engineering (Master course) *Winter 2023, Summer 2023*
- Designing Code Analyses for Large-scale Software Systems I (Master course) *Winter 2022, Winter 2021*
- Designing Code Analyses for Large-scale Software Systems II (Master course) *Summer 2022, Summer 2021*

## WORK EXPERIENCE

- Research Associate, Heinz Nixdorf Institut, Universität Paderborn *(Feb 2021 – now)*
  PI: Prof. Dr. Eric Bodden
- Project Research Assistant, Centre for Formal Design and Verification of Software *(Jul 2017 – Jul 2018)*
  Dept of Computer Science and Engineering, IIT Bombay
  PI: Prof. Supratik Chakraborty

## INTERNSHIPS

- Master's Thesis Internship, Heinz Nixdorf Institut, Universität Paderborn *(Sept – Dec 2019)*
  Supervisor: Prof. Dr. Eric Bodden
  Thesis Topic: *Call Graph Construction for Spring Framework*
- Research Internship, Dept of Computer Science and Automation, IISc Bangalore *(May – Jul 2019)*
  Supervisor: Prof. Deepak D'Souza
  Project: Data Race Detection in Android.
- Bachelor's Thesis Internship, Dept of Computer Science and Engineering, IIT Bombay *(Jul 2016 – Feb 2017)*
  Supervisor: Prof. Supratim Biswas
  Thesis Topic: *Precision Evaluation of Pointer Analysis Variants*

## EXTRACURRICULAR ACTIVITIES

- Poet of more than 275 English poems some of which have been published on web.
- Author of a blog where I share my travel experiences.
- Member of the CMI Literature Club. *(2018 – 2020)*